Question 3 Part d.)

$$A = \begin{pmatrix} 4 & -2 & 1 \\ 3 & 6 & -4 \\ 2 & 1 & 8 \end{pmatrix}$$

$$B_1 = \begin{pmatrix} 4 \\ -10 \\ 22 \end{pmatrix}$$

$AX = B_1$

$(A|B_1) =$

$$\left(\begin{array}{ccc|c} 4 & -2 & 1 & 4 \\ 3 & 6 & -4 & -10 \\ 2 & 1 & 8 & 22 \end{array}\right) \xrightarrow{-\frac{3}{4}R1+R2 \to R2}$$

$$\left(\begin{array}{ccc|c} 4 & -2 & 1 & 4 \\ 0 & +5/2 & -19/4 & -13 \\ 2 & 1 & 8 & 22 \end{array}\right) \xrightarrow{-\frac{1}{2}R1+R3 \to R3}$$

$$\left(\begin{array}{ccc|c} 4 & -2 & 1 & 4 \\ 0 & 15/2 & -19/4 & -13 \\ 0 & 2 & 15/2 & 20 \end{array}\right) \xrightarrow{-\frac{4}{15}R_2+R_3 \to R3}$$

$$\left(\begin{array}{ccc|c} 4 & -2 & 1 & 4 \\ 0 & 15/2 & -19/4 & -13 \\ 0 & 0 & 263/30 & 352/15 \end{array}\right) \xrightarrow{-\frac{13}{8}R3+R2 \Rightarrow R3}$$

$$X_3 = \frac{352}{15} \times \frac{30}{263} = \frac{704}{263}$$

$$X_2 = \frac{-13 + \frac{19}{4}\left(\frac{704}{263}\right)}{(15/2)} = \frac{-10}{263}$$

$$X_1 = \frac{4 - \frac{704}{263} + 2\left(\frac{-10}{263}\right)}{4} = \frac{82}{263}$$